

Genome-wide investigation and expression analysis suggest diverse roles of auxin-responsive *GH3* genes during development and response to different stimuli in tomato (*Solanum lycopersicum*)

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Abstract In plants, auxin-mediated responses are regulated by diverse proteins. One such class of proteins, i.e. GH3, is involved in the conjugation of IAA to amino acids and provides a negative feedback loop to control auxin homeostasis. In order to have a better understanding of the mechanism of the auxin action, 15 genes encoding GH3 members were identified using existing EST databases of tomato. Their orthologs were identified from tobacco, potato, *N. benthamiana*, pepper, and petunia. Phylogenetic analysis of *AtGH3*, *SIGH3*, and their Solanaceae orthologs provided insights into various orthologous relationships among these proteins. These genes were found to be responsive to a variety of signals including, phytohormones and environmental stresses. Analysis of AuxRE elements in their promoters showed variability in the sequence as well as number of this element. Up-regulation of only 11 *SIGH3* genes, in response to exogenous auxin, suggested possible relationship between the diversity in the sequence and number of AuxRE element with the auxin inducibility. Expression analysis of *SIGH3* genes in different vegetative and reproductive tissues/stages suggested limited or no role for most of the *SIGH3* genes at the initiation of fruit

ripening. However, up-regulation of *SIGH3-1* and -2 at the onset of fruit ripening indicates that these genes could have a role in fruit ripening. The present study characterizes *GH3* gene family of tomato and its evolutionary relationship with members of this family from other Solanaceae species and *Arabidopsis*. It could help in the identification of *GH3* genes and revelation of their function during vegetative/reproductive development stages from other Solanaceae members.

Keywords Auxin · *GH3* · Phylogenetic analysis · Solanaceae · Tomato

Introduction

The phytohormone auxin (IAA) is known to regulate numerous aspects of plant growth and development. In order to gain insights into molecular mechanism of its action, several auxin-responsive genes have been identified from many plant species, such as pea, soybean, tobacco, *Arabidopsis* and cucumber (Hagen and Guilfoyle 2002). These auxin-responsive genes belong to three major groups, including *Aux/IAA*, *Small Auxin Up RNA (SAUR)* and *Gretchen Hagen 3 (GH3)*. Presence of putative auxin-responsive elements (AuxREs, TGTCTC) in their promoters was attributed to their auxin responsiveness. These AuxREs motifs can vary either in number or sequence (TGTCCC, TGTCAC) (Hagen et al. 1991; Li et al. 1994; Ulmasov et al. 1995; Hagen and Guilfoyle 2002). Members of another gene family, namely ARF, were found to regulate the expression of these auxin-responsive genes (Guilfoyle and Hagen 2007). Since GH3 proteins are involved in maintenance of auxin homeostasis, it is imperative to have a better understanding of their function

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during various aspects of growth and development, to have a better understanding of the mechanism of auxin action in these responses.

The information on IAA-regulated molecular processes such as its perception, action, transport and biosynthesis has been generated from studies on the model dicot plant *Arabidopsis* (Woodward and Bartel 2005; Delker et al. 2008). In plants, several genes involved in intracellular auxin homeostasis have been identified. Their encoded proteins are involved in formation of auxin conjugates where conjugation between free IAA and an amino acid occurs. This process leads to decline in levels of free IAA in a cell and is catalyzed by auxin-inducible GH3 proteins. These proteins negatively control the auxin homeostasis in plants (Staswick et al. 2005). Domingo et al. (2009) reported that over expression of *OsGH3-1* caused reduction in free auxin content and resulted in dwarf transgenic plants of rice. Furthermore, identification of numerous IAA-amino acid conjugates as endogenous compounds from several plant species has also confirmed the occurrence of such mechanism operating in plants (Cohen 1982; Sitbon et al. 1993; Rampey et al. 2004; Ludwig-Muller et al. 2009; Staswick 2009; Ludwig-Muller 2011). The first GH3 gene was isolated from soybean by differential screening of auxin-responsive genes following auxin treatment (Hagen et al. 1984). Since then, several members of this gene family have been identified in a number of plant species ranging from mosses to gymnosperms to angiosperms (Terol et al. 2006; Wang et al. 2010; Okrent and Wildermuth 2011). GH3 proteins have been implicated in several developmental processes in plants. Two *Arabidopsis* GH3 proteins namely DFL1 and YDK1 were reported to negatively regulate shoot cell elongation and lateral root formation (Nakazawa et al. 2001; Takase et al. 2004). Khan and Stone (2007b) showed that *AtGH3-9* gene is involved in regulation of primary root growth in *Arabidopsis*. Chen et al. (2010) provided insights into the reaction catalyzed by the GH3 family of enzymes to modulate plant hormone action. They studied rice GH3-8 protein and reported that it requires either Mg^{2+} or Mn^{2+} for its maximal activity. Based on initial biochemical studies, sequence relationships and acyl-acid substrate preference of various GH3s, these enzymes were grouped into three families, i.e. I, jasmonate-conjugating; II, auxin- and salicylic acid-conjugating and III, benzoate-conjugating. Recently, function of GH3 proteins was reported in the maintenance of auxin levels in rice and *Physcomitrella patens* (Ding et al. 2008; Ludwig-Muller et al. 2009). A rice GH3 protein (OsGH3-8) was found to be involved in IAA conjugation (Ding et al. 2008). It is also known that several endogenous as well as exogenous factors, including auxin, abscisic acid (ABA), ethylene (C_2H_4), brassinolide (BL), gibberellins (GA), jasmonic acid (JA), salicylic acid

(SA), light and biotic/abiotic stresses modulate expression of GH3 genes in plants (Nakazawa et al. 2001; Staswick et al. 2005; Jain et al. 2006b; Jagadeeswaran et al. 2007; Khan and Stone 2007a; Park et al. 2007; Zhang et al. 2007, 2009; Domingo et al. 2009; Böttcher et al. 2010, 2011; Wang et al. 2010). A GH3-like gene, *CcGH3*, isolated from *Capsicum*, is regulated by both auxin and ethylene (Liu et al. 2005), whereas another gene, *Jasmonate resistant 1* (*Jar1*) is known to contribute to phytochrome and jasmonate signaling (Riemann and Takano 2008). *AtGH3-5* has been found to be involved in both auxin and salicylic acid signaling during pathogen infection (Zhang et al. 2007). In addition, its expression is induced by various stress conditions such as cold, salt, drought, and ABA in *Arabidopsis* (Park et al. 2007). The *AtGH3-12/PBS3* is known to regulate stress-induced salicylic acid (SA) metabolism and plant defense (Jagadeeswaran et al. 2007; Okrent et al. 2009). Some GH3 genes have been found to be up-regulated in response to cadmium in *Brassica juncea* L. (Minglin et al. 2005). Several classes of transcription factors like ARFs, bZIP, and R2R3-type MYBs have been shown to affect their expression across plant species (Tian et al. 2004; Heinekamp et al. 2004; Seo and Park 2009). Since their discovery in plants, functions of GH3 proteins are being searched *in planta*. The information which has accumulated till now indicates that their participation in plant development is highly complex.

Recently, role of a ripening-related GH3 protein in the regulation of endogenous IAA concentrations was studied in grape berries (*Vitis vinifera* L.) and it was observed that this protein might be involved in the establishment and maintenance of low IAA concentrations in ripening berries (Böttcher et al. 2010). The same group further showed involvement of another GH3 protein, i.e. GH3-2, during fruit ripening in grape berries (Böttcher et al. 2011). Further, differential expression of two GH3 genes was also reported in stage/tissue-specific manner during fruit development in non-climacteric fruits of longan (*Dimocarpus longan*) (Kuang et al. 2011). In tomato, auxin flux fluctuates twice during fruit development and ripening. Auxin accumulates to high levels prior to onset of ripening. Its level starts decreasing at the initiation of ripening and continues till more advanced stages of ripened fruits (Mapelli et al. 1978). Since auxin is known to delay fruit ripening in several crop species, including tomato, it is possible to delay ripening process by modulating the genes involved in auxin homeostasis during fruit ripening (Hewett 2006). Till now, there is no report on the broad characterization of any GH3 gene during various vegetative and reproductive developmental stages in tomato, although a few genes have been reported in various EST/cDNA databases as well as in the available microarray chips of tomato. Here, we report the identification and a

comprehensive analysis of *GH3* gene family from tomato (*Solanum lycopersicum*), a model plant for the Solanaceae family. Along with their identification, the work incorporates analysis of their genomic organization, chromosomal distribution, sequence homology, phylogenetic relationship, and responses to developmental and various external cues. Phylogenetic relationship of these *SIGH3* genes with their orthologs, identified from other Solanaceae members as well as *Arabidopsis*, was also studied. In silico analysis of their putative promoters for the presence of AuxRE elements and correlation between its presence with their auxin inducibility were also analyzed.

Materials and methods

Identification of genes and chromosomal localization

Members of *GH3* gene family in the tomato genome were identified using the TBLASTN search. In order to find out tomato *GH3* genes, 20 protein sequences of the *Arabidopsis* GH3 members and 12 protein sequences of rice GH3 members were searched among tomato sequences at tomato gene index project database (<http://compbio.dfci.harvard.edu/tgi/cgi-bin/tgi/gimain.pl?gudb=tomato>), tomato plant GDB database (<http://www.plantgdb.org>) and Solanaceae Genomics Network database (www.sgn.cornell.edu). All the obtained sequences (score ≥ 100 and e value $\leq e^{-10}$) were sorted for the unique sequences. GH3 domain search for these sequences were carried out using SMART database (<http://smart.embl-heidelberg.de/>). In total, 15 GH3 sequences were obtained and named as *Solanum lycopersicum* *GH3* (*SIGH3*) genes. For mapping and retrieving full genomic sequence, the *SIGH3* sequences were further used as query sequences for the BLASTN search against the SGN Tomato Whole Genome Scaffolds data (2.40) (<http://www.sgn.cornell.edu/tools/blast/>; The International Tomato Genome Sequencing Consortium). Simultaneously, 2-kb nucleotide sequences upstream to the start codon of these genes were extracted from the genomic sequence and were searched for the presence of various *cis*-regulatory elements using database of Plant *cis*-acting Regulatory DNA Elements (PLACE; Higo et al. 1999). Genes were assigned numbers from *SIGH3-1* to *SIGH3-15* based on their position from the top to the bottom on the tomato chromosomes 1–12. Further, the members of *SIGH3* were used as query sequences to search their orthologs in five other Solanaceae members, namely, potato, tobacco, *N. benthamiana*, pepper and *Petunia*, using DFCI-Plant Gene Index Database (<http://compbio.dfci.harvard.edu/tgi/plant.html>) and SGN Database (<http://www.sgn.cornell.edu>) and were named based on their relative homology with the *SIGH3* genes.

Phylogenetic analysis

A phylogenetic analysis was performed by aligning all the GH3 protein sequences of tomato (15) and *Arabidopsis* (20) with ClustalX 1.83 program and an un-rooted neighbor-joining phylogenetic tree was constructed (Saitou and Nei 1987; Thompson et al. 2002). Bootstrap analysis was carried out taking 1,000 replicates. The FigTree v1.2.2 software was used to view the phylogenetic tree (Rambaut 2008). Similarly, second phylogenetic tree was constructed using GH3 protein sequences from tomato and five members of Solanaceae family mentioned in the previous section.

Plant material and chemical treatment

Wild-type tomato plants (*Solanum lycopersicon* cv Pusa Ruby) were grown in a greenhouse at $28 \pm 1^\circ\text{C}$ with 16 h supplemental lighting, followed by 8 h dark. Fruits were harvested from various stages, including 8, 15, 30, 35, 40, 43, 45 and 60 days after pollination (DAP) as described previously (Kumar et al. 2011). Three fruits per stage per plant were harvested and fruits were cut into two halves before freezing. Before auxin treatment, apical portion (10 mm) of 3-day-old etiolated tomato seedlings were incubated in KPSC buffer (10 mM potassium phosphate, pH 6.0; 2% sucrose; 50 μM chloramphenicol) for 12 h to deplete endogenous auxins. The buffer was changed every 2 h. For auxin treatment, these were transferred to fresh buffer with 50 μM concentration of indole-3-acetic acid (IAA) for 3 h. Seedlings incubated in KPSC buffer without IAA for the same period were used as a control for this experiment. Ten-day-old tomato seedlings were used for chemical treatments. Tomato seedlings were immersed in different hydroponic culture solutions containing 50 μM of abscisic acid (ABA), BAP (6-benzylaminopurine), ethylene (ethe-rel), gibberellins (GA), jasmonic acid (JA; methyl jasmonate), SA (salicylic acid) and 10 μM of brassinolide (BL) for 1 and 3 h. Seedlings treated with water for the same time periods served as controls for the treatments, respectively. For stress treatments, tomato seedlings were exposed to cold (4°C), desiccation (seedlings placed on 3-mm blotting paper), or salt (200 mM NaCl) for the 2 h. To obtain leaf, root and shoot tissue, untreated seedlings were utilized. Floral tissues at flower bud and fully blossomed flower stages were also collected. Harvested tissues were frozen in liquid nitrogen immediately and stored at -80°C .

Microarray data analysis

Affymetrix GeneChip[®] tomato genome arrays were used to study fruit transcriptomes of wild type during ripening. The preparation of cDNA, labeling and hybridizations were carried out, as described earlier (Kumar et al. 2011).

GeneChip® Hybridization Oven 640 was used for hybridization while Affymetrix fluidic station model 450 was used to wash and stain the arrays with streptavidin–phycoerythrin. Scanning of the arrays was performed using GeneChip® Scanner 3000. ArrayAssist™ 5.0 data software was used to analyze the CEL files generated by GeneChip® Operating Software (GCOS). The normalization of probe signals was performed by the GCRMA method. To obtain expression values, average values of three biological replicates were used. Genes exhibiting greater than twofold difference (at p value ≤ 0.05) in their expression were considered to be differentially expressed. Fruit stage-specific comparisons between different stages of wild-type fruits were carried out and Benjamini Hochberg correction for FDR was applied. Tomato Functional Genomic Database (<http://ted.bti.cornell.edu/>) was used for annotation of the differentially expressed genes during ripening (Fei et al. 2006).

RNA isolation and QPCR analysis

Total RNA from all the tissues was extracted using Tri-reagent (Sigma) and DNase (Qiagen, Germany) treatment was given to it as per manufacturer's instruction. Primers were designed for the selected genes from 3' end of the gene using PRIMER EXPRESS version 2.0 (PE Applied Biosystems, USA) (Supplementary file S1). Specificity of each primer to its corresponding gene was checked using the Basic Local Alignment Search Tool of the NCBI. Three biological replicates were taken and for each three technical replicates were employed. The reactions were carried out in 96-well optical reaction plates (Applied Biosystems, USA). RT-PCR was performed using the ABI Prism 7000 Sequence Detection System and Software (PE Applied Biosystems, USA). To normalize sample variance, *GAPDH* gene was used as the endogenous control. For PCR reactions, ΔC_T values were obtained by subtracting C_T (cycle threshold) value of *GAPDH* from C_T value of the gene and the relative mRNA level was then calculated by formula: relative mRNA level = $2^{-\Delta\Delta C_T}$. To facilitate the profile match between the data obtained from QPCR and microarrays, fold change values obtained from QPCR were multiplied by average signal value obtained for mature green fruit in microarray experiment and bar charts were plotted using Microsoft excel.

Results

Identification and structural analysis of *SIGH3* genes

In plants, members of *GH3* genes have been shown to belong to a multigene family. In order to identify members of *GH3* gene family in tomato, *Arabidopsis* and rice *GH3*

proteins were utilized for TBLASTN search against the various open access genomic resources, including the recently available tomato genome sequence data as well as EST data present at DFCI tomato gene index and SGN databases, and a total of 15 *GH3* genes were identified (Fig. 1; Table 1; Supplementary file S2). The genes were designated as *S. lycopersicum GH3* genes (*SIGH3-1* to *-15*). Using Tomato WGS data (S12.40), complete genomic sequence for these genes were retrieved. All the *SIGH3* protein sequences were found to be complete. The sequence analysis of the 15 *SIGH3* proteins showed presence of typical GH3 domain in all the proteins. The size of these GH3 proteins varied from 495 amino acids (*SIGH3-9*) to 843 amino acids (*SIGH3-11*). A comparison of all the EST sequences with the corresponding genomic DNA sequence showed that the coding sequence of *SIGH3* genes are disrupted by one to seven introns (Table 1). The multiple-sequence alignment of all the *SIGH3* proteins, performed using ClustalX, showed significant conservation in core region as compared to the N- and C-terminal regions of these predicted proteins (Fig. 1). Pair wise analysis of these *SIGH3* protein sequences revealed that the overall similarity level ranged from as low as 32.09% between *SIGH3-3* and *SIGH3-8* to a highly similar level of ~90.43% between *SIGH3-13* and *SIGH3-14* (Supplementary file S3). Chromosome localization study of *SIGH3* genes demonstrated that all of these genes corresponded to different loci in the tomato genome. Fifteen *SIGH3* genes were found to be distributed on seven of the 12 tomato chromosomes. Five of the *SIGH3* genes were present on chromosome 10, three on chromosome 7, two each on chromosomes 1 and 2, while one each on chromosomes 5, 6 and 12 (Table 1). In silico analysis of AuxREs showed variation in their number as well as sequence. In addition, other *cis*-regulatory elements responsive to ethylene, ABA, SA, drought, and salt in the promoters of all the *SIGH3* genes were also identified (Supplementary file S4).

Identification of *SIGH3* orthologs in other Solanaceae members

Using *SIGH3* proteins as query sequences for TBLASTN search against TIGR and SGN EST databases, ten, seven, seven, three and one orthologous genes from tobacco (*Nicotiana tabacum*), potato (*Solanum tuberosum*), *Nicotiana benthamiana*, pepper (*Capsicum annum*) and petunia (*Petunia hybrida*), respectively, were identified. The details of the databases used in this study have been given in Supplementary file S5. Further, nomenclature of these genes was carried out based on their relative homology with the *SIGH3* genes and were named as *Nicotiana tabacum GH3* (*NtGH3*), *Solanum tuberosum GH3*

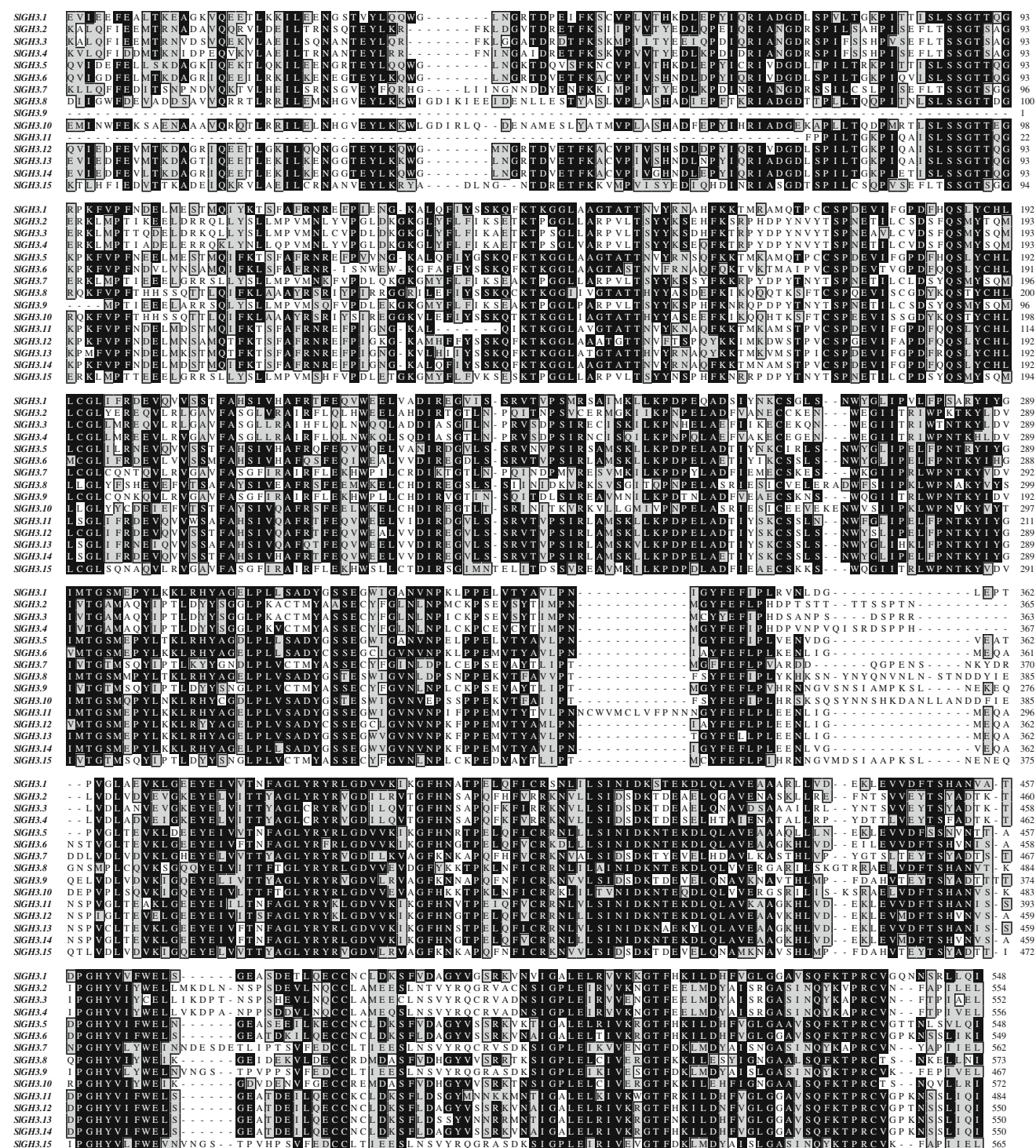


Fig. 1 Multiple alignment profile of GH3 domain of SIGH3 proteins obtained with the ClustalX program. All the sequences show high level of amino acids conservation. Gaps (dashes) have been introduced to maximize the alignments

(StGH3), *N. benthamiana* GH3 (NbGH3), *Capsicum annum* (CaGH3) and *Petunia hybrida* (PhGH3) genes, respectively. Their protein sequences along with either TIGR or SGN accession numbers are given in Supplementary file S6.

Phylogenetic analysis of SIGH3 proteins

Phylogenetic relationship between the SIGH3 protein sequences and their orthologs from *Arabidopsis* was studied by constructing a phylogenetic tree using ClustalX and

Table 1 Members of *GH3* gene family in tomato

Gene name	Scaffold WGS (S12.40)	Start	End	Protein length (aa)	Mol Wt (kDa)	pI	Intron	Chromosome	AuxRE sequence
<i>SIGH3-1</i>	Sc4323	78555183	78557768	574	64.25	5.81	3	1	1 (X)
<i>SIGH3-2</i>	Sc4323	86711873	86709859	597	67.66	5.86	2	1	3 (↑)
<i>SIGH3-3</i>	Sc3665	30537267	30532091	807	90.36	7.72	7	2	2 (↑)
<i>SIGH3-4</i>	Sc3665	48372141	48369926	599	67.57	5.91	2	2	1 (↑)
<i>SIGH3-5</i>	Sc3902	59491315	59496770	577	64.92	6.23	3	5	2 (↑)
<i>SIGH3-6</i>	Sc5732	28023080	28020038	612	68.46	6.94	5	6	4 (X)
<i>SIGH3-7</i>	Sc3685	58796583	58798815	606	68.90	5.34	2	7	0 (↑)
<i>SIGH3-8</i>	Sc3685	60182190	60178505	600	68.38	7.32	4	7	0 (X)
<i>SIGH3-9</i>	Sc4626	63429063	63430634	495	56.06	6.35	1	7	0 (↑)
<i>SIGH3-10</i>	Sc5925	2637750	2642990	602	68.78	6.92	4	10	2 (↑)
<i>SIGH3-11</i>	Sc5925	3738229	3730239	843	96.58	5.44	7	10	–
<i>SIGH3-12</i>	Sc5925	3750477	3746232	577	64.41	5.53	3	10	4 (↑)
<i>SIGH3-13</i>	Sc5925	3763423	3760899	577	64.71	6.24	3	10	1 (↑)
<i>SIGH3-14</i>	Sc5925	3904513	3901556	580	64.77	5.54	3	10	0 (X)
<i>SIGH3-15</i>	Sc4607	191139	189137	609	68.82	6.05	2	12	2 (↑)

Genes exhibiting ≥ 2 -fold induction in their expression upon auxin treatment were considered as up-regulated genes. Number present in the last column represents the frequency of the AuxRE sequence in the 2 kb upstream regulatory sequence while ↑ and X represent up-regulation or no substantial change in the expression of that gene upon auxin treatment, respectively

–, 2 kb upstream regulatory sequence for that gene could not be retrieved from tomato WGS (S12.40)

viewed in FigTree v1.2.2. Based upon the sequence homology, the combined phylogenetic tree of *Arabidopsis* and tomato showed clustering of all the GH3 proteins in three major groups (Fig. 2). Except for the class II, other two groups were composed of both AtGH3 and SIGH3 protein sequences. The group I included nine SIGH3 proteins, including SIGH3-1, -5, -6, -8, and -10 to -14 and eight AtGH3 proteins, including AtGH3-7, -8, -10 to -12, and -18 to -20. The group II included only five AtGH3 proteins (AtGH3-9, and -13 to -16). The group III included six SIGH3 proteins (SIGH3-2, -3, -4, -7, -9, and -15) along with seven AtGH3 proteins (AtGH3-1 to -6, and -17). This combined phylogenetic tree showed presence of a total of 11 sister pairs, including five SIGH3–SIGH3 and AtGH3–AtGH3 sister pairs each along with only one SIGH3–AtGH3 sister pair. Similarly, phylogenetic analysis between SIGH3 proteins and their orthologs from five Solanaceae member proteins resulted in formation of a total of three groups including I, II, and III. Out of these three groups, group I and II included mostly SIGH3 proteins, except CaGH3-11 in group II, while group III included both SIGH3 and orthologous members from other Solanaceae plants (Fig. 3; Supplementary file S6). In total, 15 sister pairs, containing three NbGH3–NtGH3, two each between SIGH3–SIGH3, SIGH3–CaGH3, and SIGH3–StGH3, along with one each between SIGH3–NtGH3, NbGH3–NbGH3, StGH3–NtGH3, NtGH3–NtGH3, NbGH3–PhGH3, and StGH3–StGH3, were formed. The maximum

number of sister pairs were found to be present in the group III while one sister pair each was present in group II. Several points were revealed by this phylogenetic analysis. First, most of the NbGH3 proteins show phylogenetic closeness to NtGH3 proteins (NbGH3-1, NbGH3-5 and NtGH3-8, NbGH3-2 and NtGH3-2, NbGH3-8, NbGH3-10 and NtGH3-10, and NbGH3-7 and NtGH3-7) (Fig. 3). Second, CaGH3-2, CaGH3-11, StGH3-5, StGH3-7, NtGH3-1, and NtGH3-3 were found to be closer to their tomato orthologs than other protein members of these plants. Third, NbGH3-2 and NtGH3-2, NbGH3-5 and NtGH3-8, NbGH3-7 and NtGH3-7, NtGH3-4 and StGH3-3, and NbGH3-3 and PhGH3-1 proteins were found to be closer to each other than to their tomato orthologs.

Expression analysis of *SIGH3* in response to various stimuli

Since these genes are involved in formation of auxin conjugates and control auxin-mediated signaling in plants, effect of exogenous auxin on the expression of *SIGH3* genes was studied at two time points (1 and 3 h) after auxin treatment. The changes in gene expression at 1 h time point were considered fast auxin responses whereas the changes at 3 h time point were considered slow auxin responses, as reported earlier (Nebenfuhr et al. 2000). It was observed that except *SIGH3-1*, -6, -8, and -14, the remaining eleven *SIGH3* genes exhibited auxin-induced

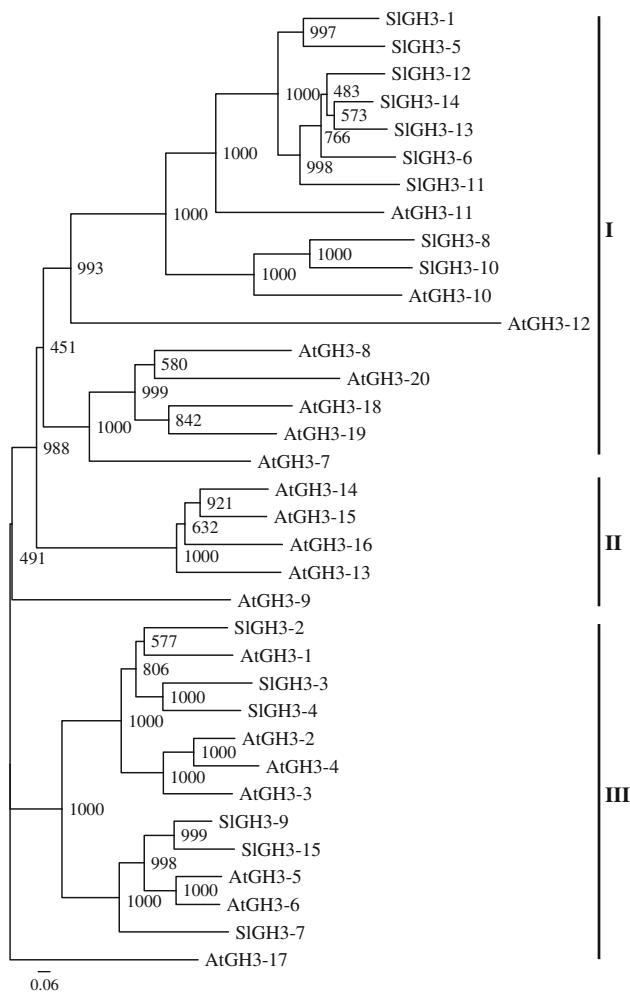


Fig. 2 Phylogenetic analysis of SIGH3 and AtGH3 proteins. The un-rooted tree was generated using the ClustalX program by neighbor-joining method and viewed in FigTree v1.2.2. Scale bar represents 0.06 amino acid substitutions per site

up-regulation at least at one of the time points. The nine *SIGH3* genes, including *SIGH3-2*, *-4*, *-5*, *-7*, *-10*, *-11*, *-12*, *-13*, and *-15* were found to be fast auxin responsive whereas *SIGH3-3* and *-9* were found to respond slowly to auxin (Table 2). The up-regulation in the expression of *SIGH3* genes was found to be more pronounced (>10-fold) for *SIGH3-2*, *-3*, *-4*, *-9*, and *-15* and at moderate levels (≤ 10 -fold) for *SIGH3-5*, *-7*, *-10*, *-11*, *-12*, and *-13* (Table 2). *GH3* genes have also been shown to respond to other phytohormones and various biotic and abiotic stresses in several plants. In addition to auxin, we also studied the effects of other phytohormones (ABA, BAP, BL, ethylene, GA, JA, and SA) and stresses (cold, salt, and desiccation) on the expression of *SIGH3* genes. Expression of *SIGH3-3* gene was found to be induced while expression of *SIGH3-8* and *SIGH3-15* was inhibited by most of the treatments (Table 2). In case of ABA and BAP treatments, no substantial alteration in the expression

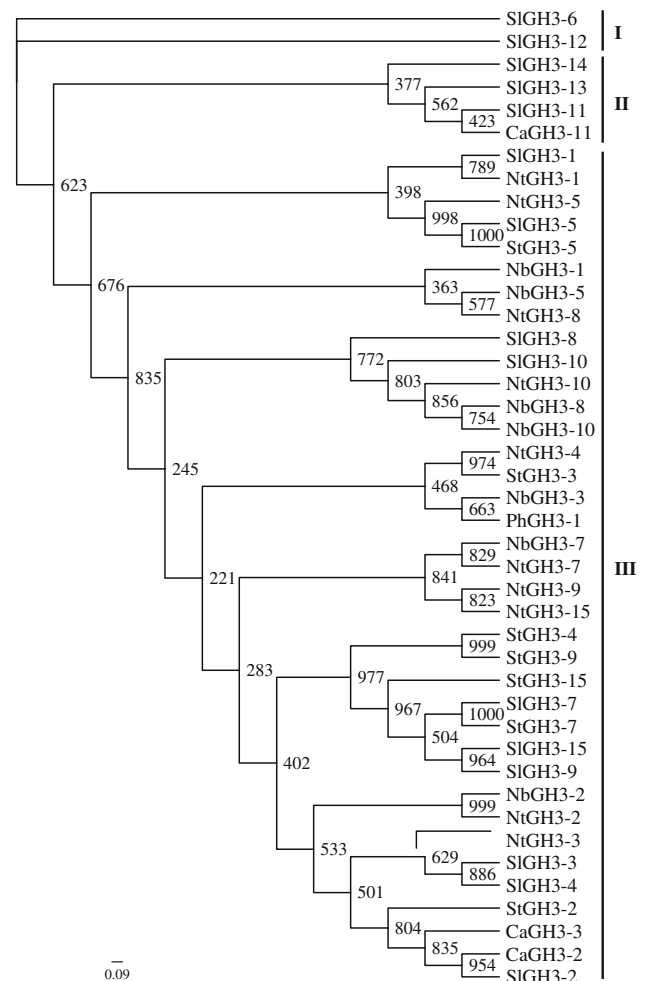


Fig. 3 Phylogenetic tree of GH3 proteins from Solanaceae. The tree depicts GH3 proteins of tomato (*SIGH3*), potato (*StGH3*), tobacco (*NtGH3*), pepper (*CaGH3*), *Nicotiana benthamiana* (*NbGH3*) and petunia (*PhGH3*). The un-rooted tree was generated using ClustalX program by neighbor-joining method and viewed in FigTree v1.2.2. In the scale bar 0.09 marks 0.09 amino acid substitutions per site

of most of the *SIGH3* genes, except *SIGH3-3*, *-4*, *-8*, *-12*, *-13*, and *-15*, was observed. Ethylene caused moderate (*SIGH3-2*, *-3*, *-4*, *-6*, *-7*, *-9*, *-12*, and *-13*) to high induction (*SIGH3-5* and *-11*) in their expression. Treatments with GA, JA and SA were found to repress the expression of most of the *SIGH3* genes except for *SIGH3-3* (Table 2). BL treatment caused induction in the expression of *SIGH3-4*, *-7*, *-8*, and *-11* whereas expression of *SIGH3-6*, *-9*, and *-12* to *-14* genes was found to be repressed (Table 2). Expression levels of several *SIGH3* genes were also found to be altered only at one of the time points under various treatments. Up-regulation in the expression of *SIGH3-4* while down-regulation of *SIGH3-12* and *-13* was observed only at early stage of ABA treatment. Expression of *SIGH3-8* was up-regulated whereas *SIGH3-13* down-regulated only at early stage under BAP treatment. *SIGH3-4*, *-7*, and *-11* were found to be slow

Table 2 Effect of various phytohormones/growth regulators and stresses on the expression *SlGH3* genes

	ABA		BAP		BL		C ₂ H ₄		GA		IAA		JA		SA		Cold		Desiccation		Salt
	1 h	3 h	1 h	3 h	1 h	3 h	1 h	3 h	1 h	3 h	1 h	3 h	1 h	3 h	1 h	3 h	2 h	2 h	2 h	2 h	
<i>SI/GH3-1</i>	-	-	-	-	-	-	-	-	R+	R++	-	-	R+	-	-	R+	R+	-	-	-	
<i>SI/GH3-2</i>	-	-	-	-	-	-	A+	A+	R++	-	A++++	A++++	-	-	R+	R+	A+	-	A+	A+	
<i>SI/GH3-3</i>	A++	A++	-	A++++	-	-	A+	A+	-	-	-	A++++	-	A++++	A++	A++++	-	A+	A++++	A++++	
<i>SI/GH3-4</i>	A+	-	-	-	-	A+	-	A+	R+	-	A++++	A++++	-	-	R+	R+	A+	A+	A+	A+	
<i>SI/GH3-5</i>	-	-	-	-	-	-	A++	A+++	-	-	A++	-	R+	-	-	R+	-	-	-	-	
<i>SI/GH3-6</i>	-	-	R+	-	-	R+	A+	A+	R+++	R++++	-	-	R+	R+	R++	R++	-	-	A++	A++	
<i>SI/GH3-7</i>	-	-	-	-	-	A+	-	A+	-	-	A+	A+	-	-	-	R++	-	-	-	-	
<i>SI/GH3-8</i>	-	-	A+	-	A++	A++++	-	-	R+	R+	-	-	R+	R+	R+	R++	R+	R+	R+	R+	
<i>SI/GH3-9</i>	-	-	-	-	R++	R+	A+	-	R++	R+++	-	-	A+++	-	-	R++++	-	-	-	-	
<i>SI/GH3-10</i>	-	-	-	-	-	-	-	-	R++	R+++	A++	A++	R+	R+	R+	R+++	-	-	-	-	
<i>SI/GH3-11</i>	-	-	-	-	-	A+	A+	A++++	-	-	A+	A++	-	-	-	-	-	-	-	-	
<i>SI/GH3-12</i>	R+	-	R+	R+	-	-	A+	A+	R++	R+++	A+	A+	R+	R+	R++	R+++	-	R+	A+	A+	
<i>SI/GH3-13</i>	R+	-	R++	-	R+	R+	A+	A+	R+	R+++	A+	-	R+	R+	R+	R+	-	-	A+	A+	
<i>SI/GH3-14</i>	-	-	R+	-	-	-	-	-	R+++	R+++	-	-	R++	R++	R+	R++	-	R+	A+	A+	
<i>SI/GH3-15</i>	R++	R+	-	-	-	-	-	-	R+++	R+++	A+	A+	R++	R++	R++++	R++++	-	R+	R+	R+	

Effects of various hormones/growth regulators was checked at 1 or 3 h after treatment. Auxin (50 μ M) treatment was given to 3-day-old etiolated tomato seedlings. In case of other hormone treatments, 10-day-old tomato seedlings were immersed in 50 μ M solutions of different chemicals, except for BL which was employed at 10 μ M concentration. The seedlings treated with water for the same time periods served as controls. The Δ ACT method was used to calculate the relative mRNA level of *SlGH3* genes after various treatments

ABA abscisic acid, *BAP* 6-benzylaminopurine, *C₂H₄* ethylene, *GA* gibberellins, *JA* jasmonic acid, *SA* salicylic acid, *BL* brassinolide, *A* activation upon hormone treatment, *R* repression upon hormone treatment

-, <2-fold; +, 2- to 5-fold; ++, 5- to 10-fold; +++, 10- to 20-fold; +++++, >20-fold

responsive while *SIGH3-12* and *-14* were found to be fast responsive to BL. Similarly some *SIGH3* genes showed fast response and some slow response to other treatments, including ethylene, GA, JA and SA treatments (Table 2). In addition to categorization of *SIGH3* genes into fast and slow responsive categories, variations were observed for both, the kinetics of transcript accumulation and the maximal induction/repression levels. Some genes which responded early continued increase in expression at late stage too (Table 2). Among different stresses, cold stress had minimum impact while salt stress had maximum impact on the expression of *SIGH3* genes (Table 2).

Expression analysis of *SIGH3* genes in vegetative and reproductive tissues/stages

Three *SIGH3* genes, including *SIGH3-2* (Les.2668.1.S1_at), *SIGH3-3* (Les.5177.1.S1_at) and *SIGH3-12* (Les.5312.1.S1_at) are represented on the Affymetrix GeneChip®. Expression of these three *SIGH3* genes was analyzed by microarray analysis and further validated by QPCR. Similar patterns in transcript accumulation for *SIGH3-2* and *-3* was observed in both QPCR and microarray analysis. In case of *SIGH3-12*, disagreement between data obtained by QPCR and microarray at *B* + 3 and *B* + 20 stages was observed (Fig. 4). In order to get better insights into expression of all the *SIGH3* genes in various vegetative and reproductive tissues/developmental stages, QPCR analysis for all the *SIGH3* genes was performed in a total of 13 vegetative and reproductive tissues/stages (Fig. 5). The genes exhibiting C_T values >36, were treated as non-expressers for the particular development stage. Expression of *GAPDH* was used as an internal control for this analysis. This analysis revealed that *SIGH3* genes are differentially expressed in various tissues/stages. The transcript levels of *SIGH3-6*, *-7*, *-8*, *-10*, and *-15* was found to be hardly detectable while *SIGH3-5*, *-9*, *-13*, and *-14* showed very less expression levels in any of the fruit stages. Expression level of *SIGH3-11* was found to be very low at most of the vegetative and reproductive tissues/stages. This gene was found to be a non-expresser at several early developmental stages (*Ft-8*, *Ft-15*, *Ft-30* and *Ft-35*) of green tomato fruit (Fig. 5). *SIGH3-1* and *-2* were the only genes exhibiting ripening-associated expression. *SIGH3-8* and *-10* exhibited only vegetative stage-specific expression while *SIGH3-7* showed highest accumulation of its transcripts in the root tissue. Several genes including, *SIGH3-3*, *-4*, *-9*, and *-15* exhibited maximum accumulation of their transcripts in shoot tissue. The expression analysis showed that all the *SIGH3* genes were found to express in more than one developmental stage/tissue in tomato (Fig. 5).

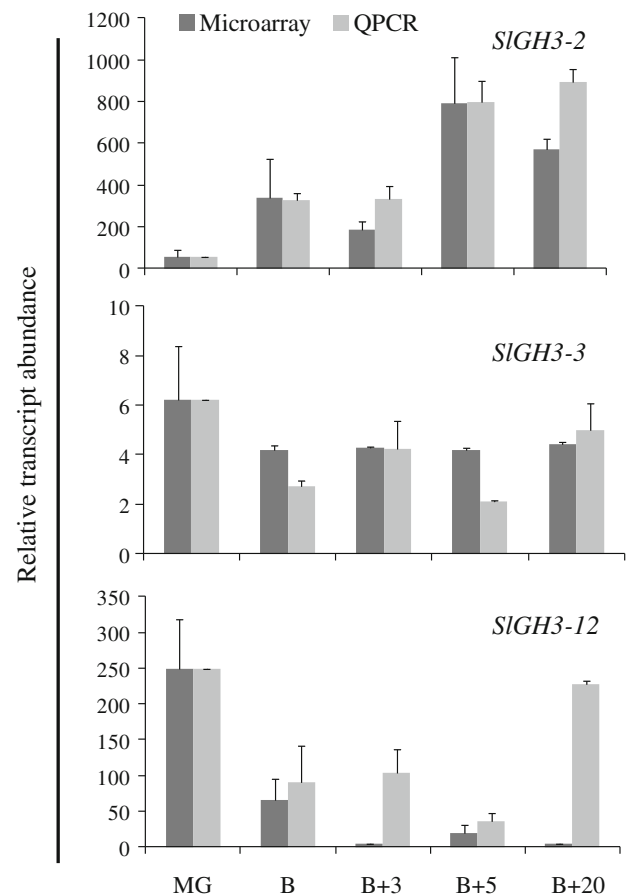


Fig. 4 QPCR results for *SIGH3-2*, *-3* and *-12* present on GeneChip® and their correlation with microarray data. The Y-axis represents raw expression values obtained from the microarray analysis in the form of relative transcript abundance. The X-axis shows five fruit developmental stages, including MG, B, B + 3, B + 5 and B + 20, in tomato. Standard error bars are shown for data obtained using both techniques. MG mature green, B breaker. B + 3, B + 5 and B + 20 represent 3, 5 and 20 days post-breaker stages of fruit ripening, respectively

Discussion

Differential screening to identify auxin-responsive genes, resulted in the identification of an auxin-induced transcript, and led to the characterization of the first *GH3* gene from soybean (Hagen et al. 1984). Later, additional *GH3* genes were also identified across diverse plant species but only recently their biochemical function was unraveled. These genes were found to provide homeostasis against the excess of auxins in plants by converting free auxins to their conjugated forms (Staswick et al. 2002, 2005). In plants, due to excess of auxins, transcription of these genes gets activated which in turn converts excess auxin to its amino acid conjugates and helps in maintenance of auxin homeostasis. Since auxins are involved in regulation of various developmental processes, such as fruit setting to ripening in

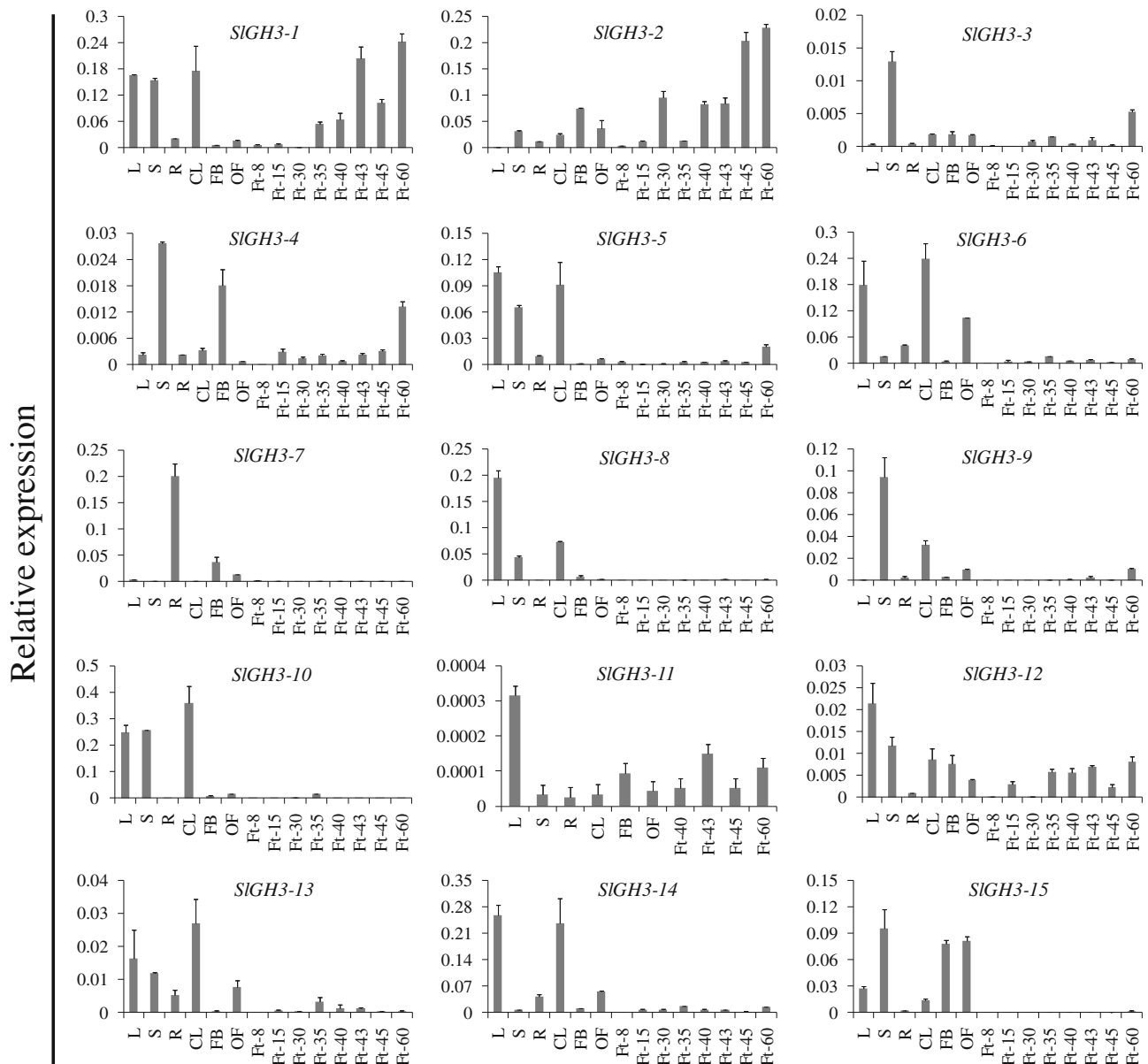


Fig. 5 QPCR analysis of 15 *SIGH3* genes in a total of 13 vegetative/reproductive tissues/stages. *SIGH3-11* gene was found to be a non-expressor at the early developmental stages (*Ft-8*, *Ft-15*, *Ft-30* and *Ft-35*) of tomato fruit. X-axis represents the different tissues/stages of development. Y-axis represents the relative mRNA levels of genes.

Expression of *GAPDH* was used as internal control and to normalize the expression of *SIGH3* genes. *L* leaf, *S* shoot, *R* root, *CL* cotyledon, *FB* flower bud, *OF* open flower. Ft is used to indicate fruit tissue while number represents days after pollination. Error bars show the standard error between three replicates performed

plants, as well as respond to several stress signals, it is necessary to study function of members of this gene family in tomato and related species so that the information could be used for improvement of agronomic characters, such as fruit quality and shelf-life and/or stress tolerance.

Identification of *GH3* genes from tomato

Initially, 20 *GH3* genes were identified from a dicot plant, *Arabidopsis* and later 12 from rice, a monocot crop plant

(Staswick et al. 2005; Jain et al. 2006b). Further, members of *GH3* gene family have been identified from various distantly related plant groups, ranging from mosses to gymnosperms to angiosperms (Terol et al. 2006; Felton et al. 2009; Okrent and Wildermuth 2011). With the availability of various genomic resources, including EST databases and recently completed tomato genome sequence databases, it has become possible to search for most of the *GH3* genes in tomato. In this study, the TBLASTN searches for tomato against the ESTs at DFCI tomato index on

TIGR and SGN databases using 20 AtGH3 and 12 OsGH3 proteins as query sequences resulted in identification of a total of 15 GH3 proteins from tomato (Table 1). The identification of lesser number of GH3 protein coding genes from tomato genome could be attributed to the relatively conserved genomes among the members of Solanaceae family, including tomato (Mueller et al. 2005). In the combined phylogenetic tree of *Arabidopsis* and tomato, all the GH3 proteins were found to cluster in three groups (Fig. 2). Absence of any *SIGH3* gene in group II indicates that in this group duplicated *AtGH3* genes were originally derived from one *AtGH3* gene and further explains for the presence of higher number of *GH3* genes in *Arabidopsis* (Lynch and Conery 2000). Similar analysis to identify *SIGH3* homologs in other Solanaceae members resulted in identification of a total of ten, seven, seven, three and one orthologous genes from tobacco, potato, *N. benthamiana*, pepper, and petunia, respectively (Supplementary file S5). These orthologous genes were named as *NtGH3*, *StGH3*, *NbGH3*, *CaGH3* and *PhGH3*, respectively, based on their homology with the *SIGH3* proteins. The reason for the identification of smaller number of *SIGH3* orthologs from other five members of the Solanaceae might be inadequacy in saturation of various EST resources. It is certain that with the increase in the number of ESTs, identification of new members of *GH3* from these plants would be possible in future. The exact number of *GH3* genes encoded by these plants will be ascertained only after sequencing of their complete genomes.

Structural analysis of *SIGH3* genes

Presence of the characteristic GH3 domain in all the *SIGH3* as well as their orthologous proteins and significant conservation in the core region as compared to the N- and C-terminal regions of all the predicted proteins show that all these proteins rightly belong to the *GH3* gene family. Presence of introns in the *SIGH3* genes follows the pattern reported earlier for the rice *GH3* genes (Jain et al. 2006b). High homologies among gene members, such as ~90.43% between *SIGH3*-13 and *SIGH3*-14, indicate that those genes might be performing essentially similar functions. However, genes with low homologies, such as *SIGH3*-3 and *SIGH3*-8 with 32.09% homology between them, indicate that these genes would be involved in distinct molecular processes (Jain et al. 2006b). To localize the identified *SIGH3* genes on the chromosomes, information on recently available tomato chromosome pseudomolecules was utilized. This study clearly demonstrated that phylogenetically related and highly homologous genes (*SIGH3*-10 to -14 on chromosome 10) are present on the same chromosome, suggesting that these genes may have a common ancestor. It is significant to mention that the

nomenclature system for *SIGH3* proteins used in the current study is based on localization of their genes on tomato chromosomes, as utilized earlier for characterization of various genome-wide gene families in plants and is widely accepted (Jain et al. 2006a, b, c; Kumar et al. 2011).

Phylogenetic relationships among orthologous GH3 proteins

In *Arabidopsis*, protein structure and specificity of GH3 proteins to adenylate plant hormones were taken as the basis for their classification (Staswick et al. 2005). In the present study, we also examined the phylogenetic relationships between the *SIGH3* protein sequences and their orthologs from *Arabidopsis*. The combined phylogenetic tree of *Arabidopsis* and tomato showed clustering of all the GH3 proteins in three major groups, exhibiting different orthologous relationships among these proteins (Fig. 2). Presence of only one *SIGH3*-*AtGH3* sister pairing (*SIGH3*-2-*AtGH3*-1) in the combined phylogenetic tree revealed limited 1:1 orthologous relationships between these *Arabidopsis* and tomato genes and indicated that these genes might have descended from a common ancestor and can have conserved functions (Kumar et al. 2011). In the second orthologous relationship, a single *AtGH3* and multiple *SIGH3* proteins (*AtGH3*-11 and *SIGH3*-1, -5, -6, -11 to -14, along with *AtGH3*-10 and *SIGH3*-8 and -10) were observed, suggesting that diversification in their conserved function might have begun in one species as a result of gene duplication events. In addition, to gain insights into the evolutionary relationship among the orthologous Solanaceae GH3 proteins, the combined phylogenetic tree resulted in formation of a total of three groups (Fig. 3). Presence of mostly *SIGH3* proteins in class I and II, except *CaGH3*-11 in class II, indicated that these proteins are phylogenetically distant to the remaining orthologous GH3 proteins. Presence of both *SIGH3* and orthologous members from other Solanaceae plants indicated phylogenetic closeness among the *SIGH3* and their orthologous proteins. Phylogenetic closeness between most of the *NbGH3* proteins to *NtGH3* proteins (*NbGH3*-1, *NbGH3*-5 and *NtGH3*-8, *NbGH3*-8, *NbGH3*-10 and *NtGH3*-10, *NbGH3*-2 and *NtGH3*-2, and *NbGH3*-7 and *NtGH3*-7), suggests that these proteins are rightly placed close to each other since both the plant species belong to same genus, i.e. *Nicotiana* (Fig. 3). Similarly, phylogenetic closeness of most of the *StGH3* proteins to *SIGH3* proteins is also as expected since both of these species belong to same genus, i.e. *Solanum*. Further, phylogenetic closeness of *CaGH3*-2, *CaGH3*-11, *StGH3*-5, *StGH3*-7, *NtGH3*-1, and *NtGH3*-3 to their tomato orthologs than other protein members of these plants, indicates that these genes might have descended from same ancestor and might be having conserved functions. Moreover,

phylogenetic relationship among a single NtGH3 and multiple SIGH3 proteins (NtGH3-3 and SIGH3-3 and SIGH3-4) suggests that due to gene duplication events, diversification in conserved function of NtGH3-3 might have begun in tomato species. Furthermore, NbGH3-2 and NtGH3-2, NbGH3-5 and NtGH3-8, NbGH3-7 and NtGH3-7, NtGH3-4 and StGH3-3, and NbGH3-3 and PhGH3-1 proteins are closer to each other than their tomato orthologs, thus providing insights into the evolutionary relationship of the SIGH3 and their orthologous Solanaceae members. Similar to 1:1 pairing between AtGH3 and SIGH3 proteins, presence of such relationship in these orthologous proteins of Solanaceae suggests that such proteins would have originated from one ancestor, before the divergence of these species (Jain et al. 2006b; Kumar et al. 2011). Further, saturation of various EST databases and completion of sequencing of other solanaceous species will yield correct and full length protein sequences that would facilitate better phylogenetic analysis of orthologous GH3 proteins in Solanaceae. Due to the well known fact that the gene numbers and gene positions are relatively conserved in almost all the Solanaceae species, the identification of *GH3* genes from tomato and their orthologs from other species will help in identification of more orthologous genes from other Solanaceous species (Mueller et al. 2005).

Regulation of *SIGH3* genes by auxin

Several published reports show that expression of several *GH3* genes is induced after auxin treatment (Staswick et al. 2005; Jain et al. 2006b; Paponov et al. 2008; Wang et al. 2010). In this study, auxin-induced up-regulation (≥ 2 -fold) of 11 out of 15 *SIGH3* genes is similar to the findings published earlier for *GH3* genes from other plants (Staswick et al. 2005; Jain et al. 2006b; Paponov et al. 2008; Wang et al. 2010). Induction in the expression of *SIGH3*-2, -4, -5, -7, -10 to -13, and -15 upon auxin treatment at 1 h time point, whereas late induction of *SIGH3*-3 and -9 suggested that these genes belonged to fast and slow responsive categories, respectively (Table 2), as described earlier (Nebenfuhr et al. 2000). The higher level of up-regulation in case of genes belonging to group III suggests that group III-mediated auxin conjugation is specifically responsive to auxin treatment (Staswick et al. 2005; Paponov et al. 2008). The variance observed in the levels of inducibility of individual *SIGH3* genes, from strong (*SIGH3*-2, -3, -4, -9, and -15) to moderate levels (*SIGH3*-5, -7, and -10 to -13) and in speed of response from fast and slow levels might be due to several reasons such as kinetics of induction, tissue-specific auxin reception, differential regulation of free auxin concentration, or differential mode of auxin-dependent transcriptional and post-transcriptional regulation (Abel and Theologis 1996; Jain et al. 2006b). Since *GH3* genes have

been classified into three families, i.e. I, jasmonate-conjugating; II, auxin- and salicylic acid-conjugating and III, benzoate-conjugating, it is quite possible that the uninduced *SIGH3* genes (*SIGH3*-1, -6, -8, and -14) would not be encoding for auxin-amido synthases and this might be an important reason for lack of their inducibility even after auxin treatment (Staswick et al. 2005). The differential response of *SIGH3* gene to auxin was found to correlate with their phylogenetic grouping (Fig. 2; Table 2). In total, six *SIGH3* genes (*SIGH3*-2, -3, -4, -7, -9, and -15) showing highest induction are more closely related (group III), whereas *SIGH3* genes with weaker or no induction are phylogenetically closer and placed in group I (Fig. 2). Further, due to the presence of several such auxin-inducible *GH3* genes in other plant species, it is suggested that controlled expression of auxin-conjugating genes may be a common mechanism that plants use to feedback-regulate the level of active auxin (Staswick et al. 2005; Domingo et al. 2009). Presence of AuxREs in the upstream regulatory sequences was found to be correlated with the auxin inducibility of several of these genes, including *SIGH3*-2, -3, -4, -5, -10, -12, -13, and -15. Up-regulation of *SIGH3*-7 and -9 by auxin even in the absence of AuxRE in their upstream regulatory sequences indicates the possibility of involvement of some other *cis*-element in such auxin-mediated up-regulation. Promoter of *SIGH3*-8 and -14 genes, lacking AuxRE in their upstream regulatory sequences, did not show any induction upon auxin treatment further supports the earlier findings that the presence and absence of AuxRE can be correlated with the inducibility or lack of it, for most of the *GH3* genes (Hagen et al. 1991; Li et al. 1994; Ulmasov et al. 1995; Guilfoyle and Hagen 2007; Wang et al. 2010).

Response of *SIGH3* to various stimuli

Besides auxin, published studies have shown that ABA, brassinolide, ethylene, GA, JA, and SA also affect the expression levels of several of the *GH3* genes across various plant species, including *Arabidopsis*, soybean, tobacco, rice and sorghum (Nakazawa et al. 2001; Sawa et al. 2002; Zhao et al. 2003; Goda et al. 2004; Tian et al. 2004; Liu et al. 2005; Jain et al. 2006b; Khan and Stone 2007a; Park et al. 2007; Zhang et al. 2009; Wang et al. 2010). Moreover, *GH3* genes have also been implicated in various biotic and abiotic stress responses, indicating cross talk between stress and hormone signaling (Jagadeeswaran et al. 2007; Nobuta et al. 2007; Matsui et al. 2008; Jain and Khurana 2009; Zhang et al. 2009; Wang et al. 2010; Okrent and Wildermuth 2011). Physiological effects of various phytohormones are known to be manifested, in part, by altered expression of genes responsive to these hormones (Nemhauser et al. 2006). In order to study whether the *SIGH3* genes are also

involved in cross talk with other phytohormone signals as well to various environment stresses in tomato, their expression in response to various external stimuli was studied. Induction of the expression of *SIGH3-3* gene under most of the treatments, except under BL, GA, and cold stress is consistent with previous reports where *GH3* genes have been found to be induced in response to several phytohormones and stress treatments (Park et al. 2007; Wang et al. 2010). No substantial change in the expression of most of the *SIGH3* genes upon ABA and BAP treatments suggests that these hormones would not be directly involved in the auxin-mediated responses controlled via these *SIGH3* genes in tomato. In case of ethylene, moderate to high induction in expression of several of these genes suggests that ethylene is also involved in GH3-mediated auxin responses as reported earlier for GH3-mediated responses in other plants (Liu et al. 2005). It was found that JA and SA repress the expression of most of the *SIGH3* genes, except *SIGH3-3*, suggesting that JA and SA can regulate expression of these genes in both positive as well as negative manner. Similar to effects of JA and SA, BL induced the expression of *SIGH3-4*, -7, -8, and -11 and reduced the expression of *SIGH3-6*, -9, and -12 to -14, suggesting that it can also regulate expression of these genes in both positive as well as negative manner (Table 2). Repression in the expression of most of the *SIGH3* genes upon GA treatment indicates that GA is involved in the negative regulation of these genes. This study further confirms earlier observations that cross talk among various phytohormone signaling operates during auxin-mediated responses in plants (DeWald et al. 1994; Tiriyaki and Staswick 2002; Khan and Stone 2007b; Park et al. 2007; Ghanashyam and Jain 2009; Zhang et al. 2009; Wang et al. 2010). Altered expression levels as well as variable responsiveness of several *SIGH3* genes, such as *SIGH3-4*, -12, -13, and -15 for ABA, *SIGH3-3*, -8, and -13 for BAP, *SIGH3-4*, -7, -8, -9, and -11 to -14 for BL, *SIGH3-4*, -5, -7, -9, -11, and -12 for ethylene, *SIGH3-1*, -2, -4, -5, -6, -9, -10, -12, -13, and -15 for GA, *SIGH3-3*, -7, -10, -12, and -15 for JA and *SIGH3-1*, -3, -5, -7, -8, -9, -10, -12, and -14 for SA treatments, between the two time points analyzed in the present study indicates complex control for availability of auxin levels (Table 2). As suggested earlier, it can be assumed that the early responding *SIGH3* genes in response to various treatments in the present study would be influenced by the particular growth regulator in a more direct way in comparison to the late responding genes (Abel and Theologis 1996). Altered expression of majority of the *SIGH3* genes in response to the three stress treatments indicates that these genes might be involved in mediating various stress responses in tomato. This study further supports the hypothesis that these genes might mediate cross talk among various phytohormone signals and environment stresses in tomato as well (Park

et al. 2007; Wang et al. 2010). Moreover, in silico study of their promoter architecture (2 kb upstream sequence from the start codon) revealed the presence of several *cis*-acting regulatory elements responsive to ethylene, ABA, SA, drought, and salt (Supplementary file S4). Since several of these genes are either induced or repressed by phytohormone signals and/or environmental stresses, these data imply that expression of these genes might be controlled by different signal pathways. Further, these results provide support to previous such studies where these *cis*-acting elements have been predicted to be involved in the regulation of *GH3* genes under such responses (Wang et al. 2010).

Study of *SIGH3* genes in vegetative and reproductive tissues/stages

Analysis of the expression of *SIGH3* genes provided more insights into their role during various vegetative and reproductive tissues/developmental stages in tomato. The transcript levels of *SIGH3-6*, -7, -8, -10, and -15 were hardly detectable in fruit tissues, indicating their limited or no role during various stages of fruit development. Expression of *SIGH3-1* and -2 was found to be up-regulated at the onset of fruit ripening, indicating that these genes would play an important role during fruit ripening in tomato (Fig. 5). These findings are in accordance with the earlier studies which showed differential regulation of *GH3* genes in stage/tissue-specific manner during fruit growth and development in the fruits of longan and grape berry (Böttcher et al. 2010; Kuang et al. 2011). Since *SIGH3-1* and -2 exhibited ripening-associated up-regulation, which correlated with the increased ethylene production in fruits, we also checked the ethylene inducibility of these two genes. It was found that only *SIGH3-2* responded to the exogenously applied ethylene, suggesting that ethylene would be involved in the regulation of its expression during fruit ripening. *SIGH3-8* and -10 exhibited only vegetative stage-specific expression, indicating that they are involved in auxin homeostasis only in vegetative tissues. High expression of *SIGH3-7* in root tissue suggests its involvement in root development in tomato (Khan and Stone 2007b). The remaining genes showed variable expression in both vegetative and reproductive tissues, indicating that these genes would be involved in auxin-mediated responses in both type of tissues (Staswick 2009; Böttcher et al. 2010, 2011; Wang et al. 2010; Okrent and Wildermuth 2011; Kuang et al. 2011).

Conclusion

In conclusion, this study provides a comprehensive gene structure analysis and expression profiling of auxin-related

GH3 genes in tomato. Further, *SIGH3-2* exhibits ripening-related expression and shows induction by auxin and ethylene and, therefore, this gene could be targeted for manipulation of the ripening process in tomato fruits. Expression analysis of *SIGH3* genes under different phytohormone treatments or stress conditions is rather intriguing and suggests that they may play a role in cross talk among various phytohormones and abiotic stress signaling pathways. Taken together, these findings suggest that members of *SIGH3* gene family are involved in the regulation of various growth/development/tissues stages and participate in various phytohormone and stress signaling pathways, yet their participation in such processes is highly complex in tomato. Since conservation in gene structure and positions among Solanaceae members is well known, the comprehensive data generated in this study will serve as a foundation for carrying out such studies and for characterization of *GH3* genes in other related species.

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